Singapore School Kinderland Secondary-2

Computer Science (0478 / 2210)

Term 2 • Academic Session 2026-27

Class Test-1: Data Storage & Data Compression (TB Pg 32–44) [ANSWER KEY]

Date: _____ **Student Name:** TEACHER ANSWER KEY & MARK SCHEME

Total Marks: 25 Marks **Marks Obtained: 25 / 25**

Teacher's Signature: _____ **Parent's Signature:** _____

Section A: Fill in the Blanks using the Technical Word Bank [5 Marks]

Word Bank: Kibibyte (KiB) • Colour Depth • Run-Length Encoding • Lossy • Sample Rate

1. In the IEC binary measurement system, exactly 1,024 Bytes is referred to as one _____ . **[Kibibyte (KiB)]**

2. The number of bits allocated to represent the unique colour of each pixel in a bitmap graphic is its _____ . **[Colour Depth]**

3. In digital audio recording, the number of sound samples captured per second is defined as the _____ . **[Sample Rate]**

4. _____ compression permanently discards imperceptible audio frequencies or minor colour variations. **[Lossy]**

5. _____ is a lossless algorithm that condenses consecutive repeated data values into frequency-value pairs. **[Run-Length Encoding]**

Section B: Write [T] for True or [F] for False [5 Marks]

No.	Technical Statement	Answer
1.	Storage manufacturers use SI metric decimal prefixes where 1 Kilobyte equals exactly 1,000 Bytes.	[T]
2.	Lossless compression is strictly required when compressing compiled software code and text documents.	[T]
3.	Decompressing a Lossy MP3 file restores the audio back to its 100% original uncompressed WAV waveform.	[F]
4.	Increasing the sample resolution of a sound recording improves sound dynamic accuracy but increases file size.	[T]
5.	A 24-bit colour depth image is capable of displaying over 16.7 million individual distinct colours.	[T]

Section C: Match File Formats with Compression Types & Attributes [4 Marks]

File Format	Compression Category & Usage	Answer
1. JPEG (.jpg)	A. Lossless image format supporting transparency, ideal for web graphics	1 → C

Section D: Multiple Choice Questions (Cambridge Paper 1 Style) [4 Marks]

1. How many Kibibytes (KiB) are contained in exactly 3 Mebibytes (MiB)?

(A) 3,072 KiB (B) 3,000 KiB (C) 1,024 KiB (D) 3,145 KiB

2. Which calculation gives the uncompressed size in Bytes of an image measuring W × H pixels with 8-bit colour depth?

(A) (W × H × 8) ÷ 8 (B) (W × H) ÷ 1024 (C) W × H × 8 (D) W × H × 64

3. What is the compressed RLE representation of the string: "B B B B B W W W R R"?

(A) 05 B 03 W 02 R (B) B5 W3 R2 (C) 5B 3W 2R 10 (D) BBBBBWWWRR

4. Why would applying RLE to the text string "C A M B R I D G E" be counter-productive?

(A) It doubles the file size because there are no consecutive repeated characters

(B) RLE cannot compress text (C) Letters are converted into binary numbers

Section E: Structured Calculation & Application Problems [7 Marks]

1. Image File Size Calculation Problem: (3.5 Marks)

A high-definition security camera captures an uncompressed image with a resolution of **1200 × 800 pixels** and a colour depth of **16 bits per pixel**.

Calculate the file size of the image in **Kibibytes (KiB)**. Show your full working steps:

- **Step 1 (Bits): 1,200 × 800 × 16 = 15,360,000 bits. [1 mark]**
- **Step 2 (Bytes): 15,360,000 ÷ 8 = 1,920,000 Bytes. [1 mark]**
- **Step 3 (KiB): 1,920,000 ÷ 1,024 = 1,875 KiB (≈ 1.83 MiB). [1.5 marks]**

2. Sound File Size Calculation Problem: (3.5 Marks)

A podcast audio recording is **20 seconds long**, recorded in **Stereo (2 channels)** at a sampling rate of **40,000 Hz** and a sample resolution of **16 bits**.

Calculate the uncompressed file size in **Bytes** and convert to **Kibibytes (KiB)**:

- **Step 1 (Bits): 40,000 × 16 × 20 × 2 (channels) = 25,600,000 bits. [1 mark]**
- **Step 2 (Bytes): 25,600,000 ÷ 8 = 3,200,000 Bytes. [1 mark]**
- **Step 3 (KiB): 3,200,000 ÷ 1,024 = 3,125 KiB (≈ 3.05 MiB). [1.5 marks]**

Teacher's Assessment Feedback & Evaluation Rubric:

Assessment Criteria	Max Marks	Marks Awarded	Teacher Qualitative Remarks & Feedback
1. Storage Unit Prefixes & Conversions	8	**8**	Strengths: **Excellent computational accuracy & formula recall** Focus Area: **Review unit prefix distinctions (KiB vs KB)** Parent's Ack: _____
2. Compression Theory & RLE Algorithms	8	**8**	
3. Image & Sound Mathematical Problems	9	**9**	
Total Marks Obtained	**25**	**25 / 25**	**Grade: [A*] [A] [B] [C]**

Evaluator Signature: **Teacher In-Charge** **Date:** _____